

Changes to the LINAC Gun High Voltage Power Supply Interlock

1.18.27.2 Rev. 1

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TABLE OF CONTENTS

1.0	Introduction.....	1
2.0	Linac Radiation Sources and Interlocks.....	1
3.0	Electron Gun HVPS Crowbar	4
4.0	Proposed Changes	4
5.0	Effects of Changes	5
6.0	Conclusion.....	5
7.0	References	5
	Revision History	6

1.0 INTRODUCTION

The electron gun produces an electron beam for operation of the CLS storage ring. Interlocks are used to shutoff the electron gun in order to prevent exposure to hazardous levels of radiation. The existing implementation of the interlock relies on signals that are generated to perform shutters open injection for top up mode operation. The nature of these interlocks could damage the electron gun high voltage power supply. The time required to repair the damage would be a very significant interruption to operation of the CLS storage ring for experimental activities. This document proposes a change that would mitigate the operational risks associated with the electron gun interlock.

2.0 LINAC RADIATION SOURCES AND INTERLOCKS

There are two radiation sources in the Linear Accelerator (LINAC); the electron gun injector and the radio frequency system.

The LINAC electron gun injector is the source of electrons for the CLS accelerators. The gun injector system uses a **High Voltage Power Supply** (HVPS) to provide a constant bias of 220 kV to a DC accelerating structure to produce a pulse of electrons with an energy of 220 keV [1]. The emission of the electron pulse is controlled by the **Gun Trigger** signals. When the triggers are enabled, electron pulses with a duration of 5 μ s are emitted from the DC accelerating structure. The repetition frequency of the electron pulses is 1 Hz. The emission of electrons from the gun poses a radiation hazard. The electron gun is interlocked to inhibit the emission of electrons.

The other radiation source in the LINAC is the Radio Frequency (RF) energy, which is used to capture and accelerate the low energy electron beam from the gun. The system consists of a 6kW **RF source**, 6 high power RF transmitters, an in-vacuum waveguide distribution network and the accelerating cavities in the LINAC [2]. The RF source uses a 15kV power supply to produce pulses of high frequency (2856 MHz) energy with a duration of 3 μ s. The pulses occur at a repetition frequency of 1 Hz and are controlled by the **RF trigger** signal. The presence of RF in the LINAC accelerating cavities poses a radiation hazard. The RF source is interlocked to prevent RF energy from entering the accelerating cavities in the LINAC.

The electron gun and accelerators are located in shielded areas of the facility to protect personnel from exposure to hazardous levels of radiation when the beam is present. These areas are set up as exclusion zones using an **Access Control and Interlock System** (ACIS) [4]. The ACIS generates permissives based on the following states:

- The state of the lockup of accelerator zones
- the state of emergency off switches in the zones
- the state of safety shutters in the LINAC-to-booster transfer line, the booster-to-storage ring transfer line, and at the front end (within the storage ring) of beamlines for shutters closed injection mode
- the state of top-up signals for shutters open injection mode.

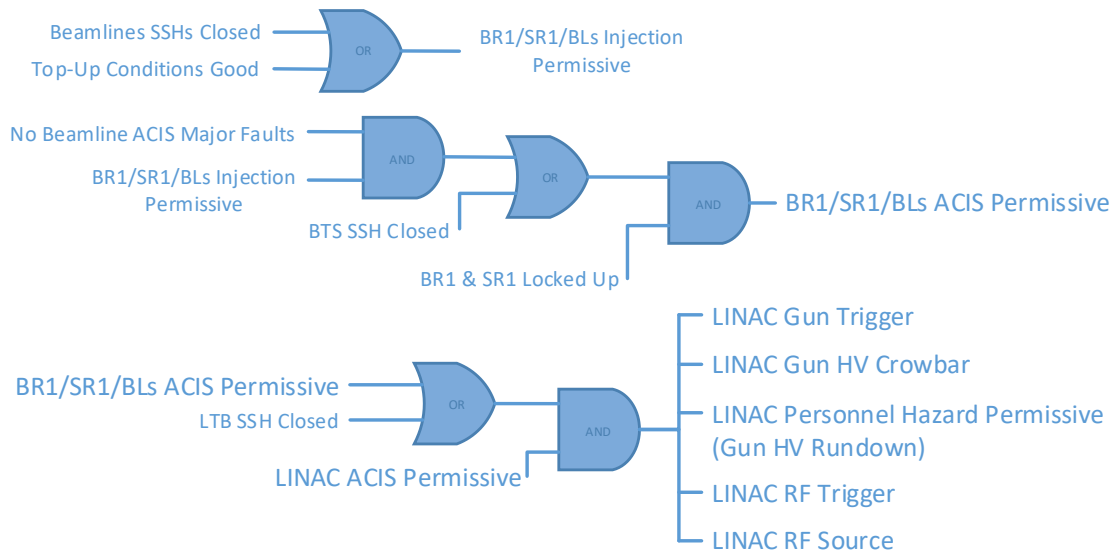


Figure 1: LINAC Gun & RF Permissives

These permissives are used to interlock the radiation sources in the LINAC. For the purpose of this discussion, the ACIS can be separated into two broad areas; the **LINAC ACIS** and the **Booster/Storage Ring/Beamline (BR/SR/BL) ACIS**. The signals from each of these areas consist of two independent chains; one governed by a safety-rated programmable logic controller (PLC), and one that uses relay-based hardware logic (HW). Multiple permissive signals are generated to interlock the LINAC radiation sources in different ways to avoid a single failure point [3] .

The LINAC **Machine Protection System (MPS) Programmable Logic Controller (PLC)** monitors the state of the LINAC and provides feedback to the operator in the control room. The state of the ACIS permissives are monitored by the PLC. Logic in the PLC is used to provide enable signals that allow the sources to be turned on. The enable signals may also be inhibited by the MPS in order to prevent damage to the equipment. This could also result in an interlock of the radiation sources.

The ACIS permissives that are used to interlock the electron gun are summarized in Table 1. The **HV Permissive** generates an interlock that places an immediate short across, or crowbars, the output of the HVPS. The **Personnel Hazard Permissive** generates an interlock that runs the HVPS output down to a low voltage before actuating the crowbar; it takes 50 seconds to run down the voltage from 220kV to 0V. The Personnel Hazard permissive is implemented in parallel with the HV Permissive such that the HV Permissive interlock takes precedence in case of an ACIS permissive. The **Gun Coax Permissive** inhibits the triggers signals that are required to generate electron pulses from the gun. This results in an immediate stoppage of the electron beam.

Gun Interlock	Response	Time	Inputs	Location
HV permissive	HV crowbar	Immediate	i. LINAC ACIS ii. BR/SR/BL ACIS iii. GUN HV ENABLE (LINAC MPS)	K9 in P1021.2-02 (ACCL/EE/GUN/PAN/0046074)
Personnel Hazard Permissive	HV rundown	50 seconds	i. LINAC ACIS ii. BR/SR/BL ACIS	K1 in Injector Status and Controls P1021.1-09 (ACCL/EE/GUN/DFR/0037353)
Gun Coax Permissive	Blocks the gun triggers	Immediate	i. AARMS Relay Interlock* ii. LINAC ACIS iii. BR/SR/BL ACIS iv. Gun trig enable (LINAC MPS)	R1021.2-09 on the bottom of the door (ACCL/EE/PROT/LOCK/0168311)

Table 1: Summary of the interlocks for the LINAC Gun HV. The combination of HV, Personnel Hazard and Gun Coax permissives results in an immediate inhibit of the electron Gun. The AARMS Relay Interlock for the Gun Coax Permissive is not currently implemented.

The permissives that are used to interlock the LINAC RF are summarized in Table 2. The **RF Source Permissive** turns off the high voltage power supply in the RF source. The **RF Trigger Permissive** blocks the trigger signal that generates the RF pulse and inhibits the low level 2856 MHz signal. Operation of the RF source presents a radiation exposure hazard to the zones that contain the LINAC. In regards to operation of the electron gun, shutting down the RF source prevents electrons from propagating outside of the LINAC zones to reach the booster ring, storage ring or beamlines.

RF Interlock	Response	Time	Inputs	Location
RF Source Permissive	Turns the HV in the RF drive amplifier off	Seconds for HV capacitor to discharge	i. LINAC ACIS ii. BR/SR/BL ACIS iii. RF_SOURCE_ENABLE (LINAC MPS)	K1 in RF Source (ACCL/EE/MOD/RF/0046057)
RF Trigger Permissive	Blocks RF trigger and inhibits low level RF	Immediate	i. LINAC ACIS ii. BR/SR/BL ACIS iii. RF_SOURCE_TRIGGER_ENABLE (LINAC MPS)	RF Trigger Board (ACCL/EE/MOD/CTL/0121510)

Table 2: Summary of the interlocks for the LINAC RF. The combination of source and trigger permissives result in an immediate inhibit of the LINAC RF.

3.0 ELECTRON GUN HVPS CROWBAR

The reason to propose a change to the existing system is in regards to the BR/SR/BL ACIS interlock that crowbars the electron gun HVPS when a fault occurs.

There is a high risk that the crowbar could damage components in the HVPS should it occur often. The HVPS provides 220kV through 25 ft. of cable to the accelerating column in the gun. This cable presents a significant amount of capacitance. When the crowbar occurs, the energy stored in the cable is discharged through the crowbar circuit and output filters of the power supply. Multiple occurrences in a short time span will result in overheating of these components resulting in failure of the electron gun. It should be noted that the original intention of the crowbar was not to routinely shut down the power supply. The interlock exists in order to provide a fast shutdown of the power supply in case of an equipment fault that could damage parts of the HV system.

Repairing the electron gun HVPS requires significant effort and cost to the facility in regards to down time. The difficulty in the repair is due to the size and age of the equipment, the requirement to filter a large amount of high voltage insulating oil, and the time required to condition the power supply back to 220kV operation. The CLS relies on the Electron Gun injector to provide electrons for the Storage Ring, and operation is suspended while the electron gun is offline for repairs. Based on our experience, the repairs could take approximately 1 month to perform; this includes time to identify and repair the failure, filter and degas the oil, and recondition the system for HV operation. This time could be longer if it is required to find replacement parts for the power supply.

Operation of the CLS accelerators in shutters open injection mode increases the risk of failure of the gun due to the permissive used for top-up injection. This permissive creates an interlock any time the current in the Storage Ring drops below a certain threshold. This is, unfortunately, a common occurrence during operation due to trips in the storage ring. It is expected that the top up injection permissive will generate a considerable amount of interlocks to the electron gun resulting in an increase in the number of crowbar events.

The preferred method to shut down the gun is with an HV rundown. The HV rundown that is triggered by the Personnel Hazard Permissive operates by slowly decreasing the voltage input to the high voltage transformer and actuating the crowbar when the output voltage has reached a low level (~10kV). The full rundown of the high voltage to zero volts takes 50 seconds. Since the stored energy is dissipated over a longer time, there is less heating in the components in the power supply. There is no risk of damaging the electron gun HVPS using this rundown.

4.0 PROPOSED CHANGES

The proposed changes are only for the ACIS interlocks of the Electron Gun. The interlocks for the RF source remain unchanged. The proposed changes are as follows:

1. Remove the BR1/SR1/BL input from the Gun HV Permissive signal wires.
2. Remove the BR1/SR1/BL input from the Gun HV Permissive signal logic in the LINAC MPS PLC.

5.0 EFFECTS OF CHANGES

The response of a LINAC ACIS interlock would remain unchanged. The electron gun would be immediately shut off using the crowbar of the HVPS. At the same time, the gun triggers would be inhibited by the Gun Coax Permissive. This response is warranted for the LINAC ACIS since the electron beam presents a radiation hazard inside the LINAC zone, even if RF is not present in the LINAC cavities. A redundant, immediate response to a LINAC ACIS interlock will ensure that any possible exposure is as low as reasonably achievable.

In the case of a BR/SR/BL ACIS interlock or storage ring top up current threshold dip, the Personnel Hazard Permissive would initiate the HV rundown which takes 50 seconds to complete. It should be noted that within the span of a few seconds, the energy is low enough that the beam would not make it out of the LINAC should the gun fire in the lower energy state. While the HVPS crowbar is faster, the HV rundown is also capable of removing a radiation source from entering the BR1/SR1/Beamlines section of the accelerators. The gun triggers would still be inhibited and immediately block electron emission from the gun. The RF source and RF triggers are also immediately inhibited due to the interlock. The absence of RF in the LINAC prevents the electron beam from propagating beyond the LINAC zone. Despite the slower response of the HV rundown, the simultaneous inhibit of the gun triggers and the LINAC RF continue to provide an immediate response to prevent electron beam from reaching the BR/SR/BL zone.

6.0 CONCLUSION

There is a risk that the existing implementation of the gun interlock, which crowbars the output of the HVPS, could damage the electron gun HVPS. The risk for potential down time will become more severe for top up mode operation since it is expected that the interlock will occur more frequently. The proposed change would rely on the HV rundown method of disabling the electron gun for a BR/SR/BL ACIS interlock, which would decrease the risk of electron gun failure. The immediate inhibit of the gun triggers and LINAC RF maintain redundant means of ensuring that the electron beam does not propagate beyond the LINAC zones.

7.0 REFERENCES

- [1] "13 MeV Injection Linear Accelerator Model HRC-735 Instruction and Maintenance Manual", Haimson Research Corporation, September 1980
- [2] "LINAC Setup Manual", 1.9.36.1
- [3] "LINAC Access Control and Interlock System (ACIS) Design Manual", 1.9.52.1
- [4] "LINAC ACIS Upgrade Hazard and Risk Analysis", 11.18.52.1
- [5] "Booster/Storage Ring/Beamlines Access Control and Interlock System (ACIS) PLC Component Manual", 7.9.39.4
- [6] "CLSI Safety Report", 11.18.40.2

REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	2018-Jun-12	Original Draft.	Bud Fogal
B	2018-Jul-18	Updated from comments. Added permissive flow chart.	Bud Fogal
0	2018-Jul-27	Issued for use.	Bud Fogal
1	2018-Aug-09	This document was incorrectly numbered as 1.18.27.1; which was already assigned to another document. The required corrections have been made and this document has been renumbered as 1.18.27.2. Issued for use.	Bud Fogal